SET-I

ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Course: **B. Tech Biotechnology** Semester: **VII**th

Name of the Paper **Enzyme Technology** Paper Code: **TBT-702c**

Time: 3 Hours Maximum Marks: 100

Note:- (i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Question 1		**CO**
(a)	1. Define enzymes and describe their general properties. (5) 2. Explain the principle of ion-exchange chromatography in enzyme purification. (5)	
(b)	**1.** Explain the structure of an enzyme with reference to its active site and substrate specificity. (5) 2. Describe the mechanism of Lock and Key hypothesis in detail. (5)	**CO1**
(c)	**1.** What are enzyme units? Explain the concept of specific activity. (5) 2. Describe in detail the technique of High pressure Liquid Chromatography. (5)	
Question 2		
(a)	Derive the Michaelis-Menten equation and explain the significance of K_m and V_{max}. Discuss how these parameters are experimentally determined. (10)	
(b)	Compare the Lineweaver-Burk, Eadie-Hofstee, and Hanes-Woolf plots for analyzing enzyme kinetics. What are the pros and cons of each method? (10)	**CO2**
(c)	1. A reaction has an initial velocity of 100 μmol/min when the substrate concentration is 20 mM. Given V_{max} is 200 μmol/min and K_m is 10 mM, Calculate the velocity at a substrate concentration of 50 mM. (4) 2. Discuss the concept of 0^{th} order and 1^{st} order enzyme kinetics with diagrams. (6)	
Question 3		
(a)	Explain in detail the positive and negative regulation of enzymes.	
(b)	What do you mean by enzyme regulation? Explain the various mechanisms of enzyme regulations in detail. (10)	**CO3**
(c)	Differentiate between competitive and non-competitive inhibition with diagrams. (10)	
Question 4		
(a)	1. Explain the role of enzyme in food industry with examples. (5) 2. Analyze the use of enzymes in environmental biotechnology. (5)	
(b)	1. Define extremophilic enzymes and discuss their applications. (5) 2. Explain the concept of artificial enzymes in detail with examples. (5)	**CO4**
(c)	Define proteases and explain their classification based on their mechanism of action and pH preferences. Discuss their industrial applications with examples, focusing on their roles in food processing, pharmaceuticals, and detergent production. (10)	
Question 5		
(a)	What do you mean by enzyme engineering? Discuss the various strategies employed in enzyme engineering to improve stability, activity, and specificity? Discuss these approaches with examples from industrial applications. (10)	
(b)	Evaluate the importance of metagenomics in discovering novel enzymes from unculturable microorganisms. How does this approach compare with traditional methods? (10)	**CO5**
(c)	Describe the various methods of enzyme immobilization, including adsorption, covalent bonding, entrapment, and encapsulation. Compare their advantages and disadvantages by providing relevant examples. (10)	